

STUDENT MANAGEMENT SYSTEM

1. PROJECT OVERVIEW

This Student Management System is a console-based application that demonstrates core CRUD (Create, Read, Update, Delete) operations. It acts as an essential foundation for a full Grade Sheet system by setting up file persistence, ensuring that student records are securely saved to a hard drive rather than disappearing when the program closes.

2. CORE C++ TOPICS USED

Object-Oriented Programming (OOP): Encapsulates student data (rollno, name, age) and behaviors (addStudent(), displayStudent()) inside a single Student class.

Binary File Handling (fstream): Uses ios::binary with .write() and .read() to save and load data as compact, secure byte streams (student.dat).

Stream Cleansing (cin.ignore()): Clears the input buffer to prevent cin.getline() from skipping inputs during sequential entries.

Control Flow: Uses a do-while loop paired with a switch-case block to drive an interactive, user-friendly menu interface.

3. CODE ARCHITECTURE & WORKFLOW

Data Layout & Representation

By wrapping data into a class, the program treats a student's entire profile as a single block of memory, making it easy to pass, save, and retrieve as a uniform unit.

Storage & Retrieval Mechanics

Data Persistence: Records are saved to student.dat in binary mode to prevent text formatting issues and protect data integrity.

Appending Records: Using ios::app allows the program to add new student data to the end of the file without overwriting old entries.

Searching: The program runs a while(file.read(...)) loop, scanning the file sequentially until it finds an object matching the requested rollno.

File-Based Deletion Strategy

Since binary files cannot easily be edited in place, the program uses a two-file swapping mechanism:

It reads student.dat line by line.

Every record that does not match the target roll number is copied to a new file named temp.dat.

The target record is skipped (deleted).

The program deletes student.dat and renames temp.dat to student.dat.

4. INTERNSHIP VALUE & NEXT STEPS

This project proves your understanding of low-level file manipulation in C++. To turn this into a true Grade Sheet Management System, you can easily expand it by adding a marks array (int marks[5]), a GPA calculation function, and an "Update Marks" feature.

CODE:

```
#include <iostream>
```

```
#include <fstream>
```

```
using namespace std;
```

```
class Student
```

```
{
```

```
public:
```

```
    char name[50];
```

```
    int rollno;
```

```
    int age;
```

```
    void addStudent()
```

```
    {
```

```
        cin.ignore();
```

```
        cout << "\nEnter student name:";
```

```
        cin.getline(name, 50);
```

```
        cout << "Enter student Roll Number:";
```

```
        cin >> rollno;
```

```
        cout << "Enter student age:";
```

```
        cin >> age;
```

```
    }
```

```

void displayStudent()
{
    cout << "\n=====";
    cout << "\nStudent Name:" << name;
    cout << "\nStudent Roll Number:" << rollno;
    cout << "\nStudent Age:" << age;
    cout << "\n=====\\n";
}

};

int main()
{
    Student s;
    int choice;

    fstream file;

    do
    {
        cout << "\n=====STUDENT MANAGEMENT SYSTEM=====\\n";
        cout << "\\n1.Add student details\\n";
        cout << "\\n2.Display student details\\n";
        cout << "\\n3.Search student details\\n";
        cout << "\\n4.Delete student details\\n";
        cout << "\\n5.Exit\\n";

        cout << "\\nEnter choice:";
    }

```

```
cin >> choice;
```

```
switch(choice)
```

```
{
```

```
case 1:
```

```
    file.open("student.txt", ios::app);
```

```
    s.addStudent();
```

```
    file << s.name << endl;
```

```
    file << s.rollno << endl;
```

```
    file << s.age << endl;
```

```
    file.close();
```

```
    cout << "\nStudent details added successfully.\n";
```

```
    break;
```

```
case 2:
```

```
    file.open("student.txt", ios::in);
```

```
    while(file.getline(s.name, 50))
```

```
{
```

```
file >> s.rollno;
```

```
file >> s.age;
```

```
file.ignore();
```

```
s.displayStudent();
```

```
}
```

```
file.close();
```

```
break;
```

```
case 3:
```

```
{
```

```
int r;
```

```
int found = 0;
```

```
cout << "\nEnter Roll Number to Search:";
```

```
cin >> r;
```

```
file.open("student.txt", ios::in);
```

```
while(file.getline(s.name, 50))
```

```
{
```

```
file >> s.rollno;
```

```
file >> s.age;
```

```
file.ignore();
```

```
if(s.rollno == r)
```

```
{
```

```
    s.displayStudent();
```

```
    found = 1;
```

```
}
```

```
}
```

```
file.close();
```

```
if(found == 0)
```

```
{
```

```
    cout << "\nStudent Not Found!\n";
```

```
}
```

```
break;
```

```
}
```

```
case 4:
```

```
{
```

```
    int r;
```

```
    int found = 0;
```

```
    cout << "\nEnter Roll Number to Delete:";
```

```
    cin >> r;
```

```
fstream temp;
```

```
file.open("student.txt", ios::in);
```

```
temp.open("temp.txt", ios::out);
```

```
while(file.getline(s.name, 50))
```

```
{
```

```
    file >> s.rollno;
```

```
    file >> s.age;
```

```
    file.ignore();
```

```
    if(s.rollno != r)
```

```
    {
```

```
        temp << s.name << endl;
```

```
        temp << s.rollno << endl;
```

```
        temp << s.age << endl;
```

```
    }
```

```
    else
```

```
    {
```

```
        found = 1;
```

```
    }
```

```
}
```

```
file.close();
```

```
temp.close();
```



```
remove("student.txt");  
rename("temp.txt", "student.txt");
```

```
if(found == 1)  
{  
    cout << "\nStudent Deleted Successfully!\n";  
}  
else  
{  
    cout << "\nStudent Not Found!\n";  
}
```

```
break;  
}
```

```
case 5:
```

```
    cout << "\nProgram Ended!\n";  
    break;
```

```
default:
```

```
    cout << "\nInvalid Choice!\n";  
}
```

```
} while(choice != 5);
```

```
return 0;
```

```
}
```

OUTPUT:

=====STUDENT MANAGEMENT SYSTEM=====

- 1.Add student details
- 2.Display student details
- 3.Search student details
- 4.Delete student details
- 5.Exit

Enter choice:1

Enter student name:lasya

Enter student Roll Number:1

Enter student age:18

Student details added successfully.

=====STUDENT MANAGEMENT SYSTEM=====

- 1.Add student details
- 2.Display student details
- 3.Search student details
- 4.Delete student details
- 5.Exit

Enter choice:1

Enter student name:Bharathi

Enter student Roll Number:2

Enter student age:17

Student details added successfully.

=====STUDENT MANAGEMENT SYSTEM=====

- 1.Add student details
- 2.Display student details
- 3.Search student details
- 4.Delete student details
- 5.Exit

Enter choice:1

Enter student name:Honey

Enter student Roll Number:3

Enter student age:17

Student details added successfully.

=====STUDENT MANAGEMENT SYSTEM=====

- 1.Add student details
- 2.Display student details
- 3.Search student details
- 4.Delete student details
- 5.Exit

Enter choice:5

Program Ended!

